

CASE STUDY ON THE ANALYSIS OF SDLC MODELS APPLIED TO A COLLEGE MANAGEMENT SYSTEM

Aim

DLC mTo analyze Sodels used in developing a College Management System and determine the most suitable model.

Objective

- Study different SDLC models.
- Compare their advantages and limitations.
- Analyze a College Management System.
- Recommend the best SDLC model.

Overview

A College Management System is an integrated software solution that manages admissions, student records, attendance, examinations, fee collection, library services, timetable preparation, faculty information and administrative activities. It provides a centralized database that allows authorized users to access information quickly and securely.

Problem Statement

Many colleges still depend on manual registers or isolated software applications for different departments. This creates duplication of records, delays in processing, poor communication, inaccurate reports and increased paperwork. Students often experience delays in accessing attendance, examination results and fee information. Faculty members spend significant time maintaining records manually, while administrators find it difficult to generate consolidated reports. Therefore, a centralized College Management System is required to automate academic and administrative operations, improve efficiency, reduce manual errors, ensure data security and provide real-time access to information.

Existing System

The existing system is partially computerized and largely dependent on manual work. Each department maintains separate records, making data sharing difficult. Updating information requires repetitive work, and report generation consumes considerable time. Manual calculations may lead to errors, while physical documents are difficult to store and retrieve.

Proposed System

The proposed system integrates every department into one application. Students can register, view attendance, check examination results and monitor fee status. Faculty members can upload attendance, marks and assignments, while administrators manage admissions, departments, examinations and reports. The system improves communication, accuracy, transparency and overall institutional productivity.

Functional Requirements

1. Student Module:
 - Registration
 - Login
 - Attendance
 - Results
 - Fee Status
 - Course Enrollment.
2. Faculty Module:
 - Attendance Entry
 - Marks Upload
 - Assignment Management
 - Timetable.
3. Administration Module:
 - Admissions
 - Department Management
 - Fee Collection
 - Examination Management
 - Reports.
4. Library Module:
 - Book Issue/Return
 - Fine Calculation
 - Inventory.

Non-Functional Requirements

- Security
- Reliability
- Availability
- Scalability
- Maintainability
- Performance
- backup and recovery
- user-friendly interface and data integrity.

Advantages

- Reduces paperwork
- minimizes human errors
- provides faster report generation
- improves communication
- centralizes data
- enhances security
- saves time and supports better decision-making.

Conclusion

Agile is the most suitable SDLC model because it supports iterative development, continuous feedback, frequent updates and easy adaptation to changing academic requirements. Combined with DevOps, it provides reliable deployment and efficient maintenance.